Iodine Vacancy Sites and Migration Paths in Bulk γ-$CsPbI_3$

Progress report

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October 2026

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Outline

1. Motivation
2. Literature and gap
3. Methods and why
4. Results so far
5. Next steps

Motivation

Why iodine vacancies?

Halide perovskites: electronically tolerant, ionically weak

- Most native defects leave no deep traps[7]
- But iodine vacancies form easily and move at room temperature[1,7]

Moving vacancies cause

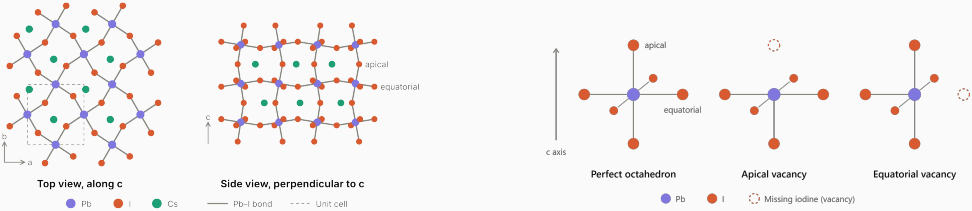
- J–V hysteresis and drifting response[7]
- Ion pile-up and reactions at contacts
- Loss of stability: γ-$CsPbI_3 \rightarrow$ yellow δ phase, starting at surface iodide defects[8]

Goal of the project
Migration barriers of the iodine vacancy in γ-$CsPbI_3$:

1. every distinct hop in the bulk (this talk)
2. layer by layer below the CsI-terminated (001) surface (later)

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [7] N. Xu *et al.*, Point defects in metal halide perovskites, Nat. Rev. Mater. (2025). [8] Z. Wylie *et al.*, ACS Energy Lett. 9, 4378 (2024).

γ-$CsPbI_3$: tilted octahedra make two iodine sites



Space group	Pnma (No. 62), 20 atoms per cell[4]
Apical I, **4c**	links octahedra along *c*, in the CsI layers; Pb–I–Pb 155°
Equatorial I, **8d**	links octahedra in the *ab* plane, in the PbI_2 layers; Pb–I–Pb 148°

The two sites differ only through the octahedral tilt (in the cubic phase they are equivalent).

[4] R. J. Sutton *et al.*, ACS Energy Lett. 3, 1787 (2018). Angles and figures: this work (tight PBEsol relaxation).

Literature and gap

Key papers

Paper	What it did	What it leaves open
A Arber 2025[1] Chem. Mater. (anchor)	Bulk γ-$CsPbI_3$, PBEsol, 2×2×2 cell. Barriers 0.34 eV (8d–8d), 0.35 eV (8d–4c), 0.77 eV (4c–4c)	One barrier per path *type*; charge state not stated; no surface
B Biega 2021[2] J. Phys. Energy	Cubic $CsPbBr_3$ slabs: long axial jump about half the bulk barrier at the surface	Long jump only; no tilt; thin, partly frozen slab never returns to bulk
C Ahmad 2024[3] Phys. Rev. Mater.	$CsPbI_3$, 17-layer slab: vacancy formation energies vs depth	No barriers; equatorial sites only; slab interior off from own bulk by 0.05–0.66 eV
D Zhang 2025[20] Appl. Surf. Sci.	Bulk γ-$CsPbI_3$ under strain, PBEsol: two paths, 0.51 eV (edge, towards apical) and 0.54 eV (in-plane); rectangular cell (PBEsol, PBE, HSE06)	Charge state; site energies; only two paths

Gap: no iodine-vacancy barriers in γ-$CsPbI_3$ resolved by hop and by depth, in a slab whose interior reproduces the bulk.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [2] R.-I. Biega, L. Leppert, J. Phys.: Energy 3, 034017 (2021). [3] B. Ahmad, M. S. R. Limon, Z. Ahmad, Phys. Rev. Mater. 8, 125402 (2024). [20] J. Zhang *et al.*, Appl. Surf. Sci. 679, 161235 (2025).

Gaps and research questions

	Gap	Research question
G1	Bulk barriers known per path type only; charge state unstated[1]	**RQ1** Which site does the vacancy prefer, in each charge state? **RQ2** Bulk barriers, and how the number and direction of hops shape transport (ab plane vs c)?
G2	Surface effect known for one jump in cubic $CsPbBr_3$ only[2]	**RQ3–4** Barriers and drift direction layer by layer below the surface
G3	Neither surface study recovers its own bulk[2,3]	**RQ5** Does the slab middle reproduce the bulk?

Scope: iodine vacancy only; charge $+1$ and neutral; CsI termination.
Interstitials, PbI_2 termination, formation energies vs μ_I and E_F, ML potentials: *out of scope for now*.

This talk: G1 (RQ1 done for $+1$, nearly done for neutral; RQ2 set up).

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [2] R.-I. Biega, L. Leppert, J. Phys.: Energy 3, 034017 (2021).
[3] B. Ahmad, M. S. R. Limon, Z. Ahmad, Phys. Rev. Mater. 8, 125402 (2024).

Methods and why

Workflow

Step	Calculation	Purpose	Status
1	Unit cell: convergence, tight relaxation	Reference lattice	**done**
2	Perfect 2×2×2 supercell (160 atoms)	Energy reference	**done**
3	4c and 8d vacancy, +1 and 0	Site preference (RQ1)	3/4 **done**, 1 **running**
4	NEB, 10 distinct bulk hops	Bulk barriers (RQ2)	inputs ready, next
5	CsI-terminated (001) slab	Bulk-like slab middle	next
6	Vacancies at increasing depth	Site energy vs depth	next
7	NEB at increasing depth	Barriers vs depth	next

Steps 1–4 are the bulk reference that every slab result is compared against.

Settings and why

Choice	Value	Why
Code	Quantum ESPRESSO 6.5[10]	One version for all production runs
Functional	PBEsol[9]	Good solid-state lattices; **same as paper A**[1]
Spin–orbit	Not included	Mainly shifts the Pb-p band[12]; check later
Dispersion	None	Same as paper A[1]
Pseudopotentials	SSSP 1.3 PBEsol efficiency[11]	Verified library
Cutoff	60 Ry (480 Ry density)	0.86 meV/atom from 80 Ry
k-mesh	4×4×3 cell; 2×2×2 supercell	0.23 meV/atom from 6×6×4; same density
Forces	largest < 0.01 eV/Å	Same as paper A[1]

Guiding rule: match paper A where possible, so any difference in results is physics, not settings.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [9] J. P. Perdew *et al.*, Phys. Rev. Lett. 100, 136406 (2008). [10] P. Giannozzi *et al.*, J. Phys.: Condens. Matter 21, 395502 (2009); 29, 465901 (2017). [11] G. Prandini *et al.*, npj Comput. Mater. 4, 72 (2018). [12] J. Even *et al.*, J. Phys. Chem. Lett. 4, 2999 (2013). Cutoff and k-mesh: this work (next slide).

Convergence tests (20-atom cell)

k-mesh (50 Ry)	meV/atom vs 6×6×4
2×2×1	45.46
3×3×2	2.52
4×4×3	**0.23**
5×5×4	0.004

Cutoff	meV/atom vs 80 Ry	P (kbar)
40 Ry	5.83	−1.89
50 Ry	3.29	−1.57
60 Ry	**0.86**	−1.48
70 Ry	0.40	−1.35
80 Ry	0	−1.39

Both choices are well below 1 meV/atom, far smaller than the energy differences we compare (tens of meV and more).

All values: this work (Quantum ESPRESSO 7.6, PBEsol, SSSP).

Unit-cell relaxation

	a (Å)	b (Å)	c (Å)	Volume (Å^3)
Experiment, 293 K[4]	8.577	8.856	12.472	947.3
Soft relaxation	8.378	8.933	12.352	924.4
Tight relaxation	**8.362**	**8.961**	**12.350**	**925.4**

- **Why a tight second relaxation?** The soft result still had a and b drifting by 0.2–0.3%. Tight limits: 0.01 eV/Å, 0.2 kbar.
- The cell is very soft against exchanging a and b (only 1.9 meV per cell over the whole tight run).
- Volume 2.3% below room-temperature experiment: expected for a 0 K PBEsol relaxation (no thermal expansion)[5].

[4] R. J. Sutton *et al.*, ACS Energy Lett. 3, 1787 (2018). [5] J. Kaczkowski, I. Płowaś-Korus, J. Phys. Chem. Lett. 12, 6613 (2021). Relaxations: this work.

Defect supercell: choices and why

Choice	Why
2×2×2 supercell, 160 atoms	Same as paper A[1]; vacancy and its images about 17–25 Å apart (1 vacancy per 96 I sites)
Cell fixed at the bulk lattice	Models one dilute defect inside a large host whose lattice the defect cannot change[13]
Charge +1 first	Closed shell; stable charge state of V_I over most of the gap[3,19]; least sensitive to spin–orbit
Neutral as well, spin-polarised	One unpaired electron; tests which charge state paper A used[1]
k-mesh kept at 2×2×2	Test with 2×2×1: forces 4× larger, 1.8 meV/atom shift → rejected
No charge correction (yet)	Cancels in site differences and barriers (same charge at both ends); needed only for formation energies[13]

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [3] B. Ahmad, M. S. R. Limon, Z. Ahmad, Phys. Rev. Mater. 8, 125402 (2024). [13] C. Freysoldt *et al.*, Rev. Mod. Phys. 86, 253 (2014). [19] Y. Huang, W.-J. Yin, Y. He, J. Phys. Chem. C 122, 1345 (2018). *k*-mesh test: this work.

Results so far

Perfect supercell: a consistency check

Total energy	−65326.88713687 Ry
Valence band maximum	3.4105 eV
Relaxation	converged on the **first** SCF, 0 BFGS steps
Total force (all atoms)	0.058 eV/Å

- The supercell built from the tight unit cell needed no relaxation.
- $\Rightarrow$ the 4×4×3 unit-cell mesh and the 2×2×2 supercell mesh agree on forces.
- Its energy and VBM are the references for the vacancy cells.

This work (job 217406).

RQ1: site preference depends on the charge state

	$E(4c)$ (Ry)	$E(8d)$ (Ry)	$\Delta E = E(8d) - E(4c)$
V_I^{+1}	-64954.16973158	-64954.16980915	-1 **meV**
V_I^{0}	-64953.80080990^{*}	-64953.79815909	$\geq +36$ **meV***
Paper A[1] †			≈ -30 meV
Zhang 2025[20] ‡			$\approx +15$ meV

* 4c neutral still relaxing (energy still falling), so the gap can only grow. † Inferred from Fig. S1; ‡ from Fig. 6c (edge path towards apical, if it runs 8d→4c); charge states not stated.

- $+1$: the two sites are **degenerate** (1 meV, below our accuracy).
- Neutral: clear preference for the **apical** site, opposite in sign to paper A.
- Neither charge state reproduces paper A's 0.03 eV; the other PBEsol study (rectangular cell) leans the same way as our neutral result.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). Not reported as a site energy. Inferred from its Fig. S1: the 8d→4c profile ends ≈0.03 eV above its start, and the caption gives forward and reverse barriers differing by ≈0.03 eV; read off a plot, ±0.01 eV.
[20] J. Zhang *et al.*, Appl. Surf. Sci. 679, 161235 (2025). Our energies: this work.

The neutral vacancy is a shallow donor (in PBEsol)

- Neutral V_{I} carries one extra electron. Two possibilities:
  - localised at the vacancy $\rightarrow$ magnetic moment about $1\,\mu_B$
  - spread over the crystal (Pb 6p conduction band) $\rightarrow$ about $0\,\mu_B$
- **Found:** $0.00\,\mu_B$ **in both cells, throughout the relaxation**; no Pb–Pb pairing.
- $\Rightarrow$ shallow donor: the electron is delocalised, consistent with earlier work on $CsPbI_3$[19,20].

Caveat

Semilocal functionals (PBEsol) tend to over-delocalise electrons (delocalisation error)[14]. A hybrid-functional check would confirm or overturn this.

Neutral 4c relaxation: energy still falls 0.4–1.0 meV per step while the moment stays 0 $\rightarrow$ a slow *structural* (tilt) relaxation, not electron localisation.

[14] A. J. Cohen, P. Mori-Sánchez, W. Yang, Science 321, 792 (2008). [19] Y. Huang, W.-J. Yin, Y. He, J. Phys. Chem. C 122, 1345 (2018). [20] J. Zhang *et al.*, Appl. Surf. Sci. 679, 161235 (2025). Magnetisation: this work.

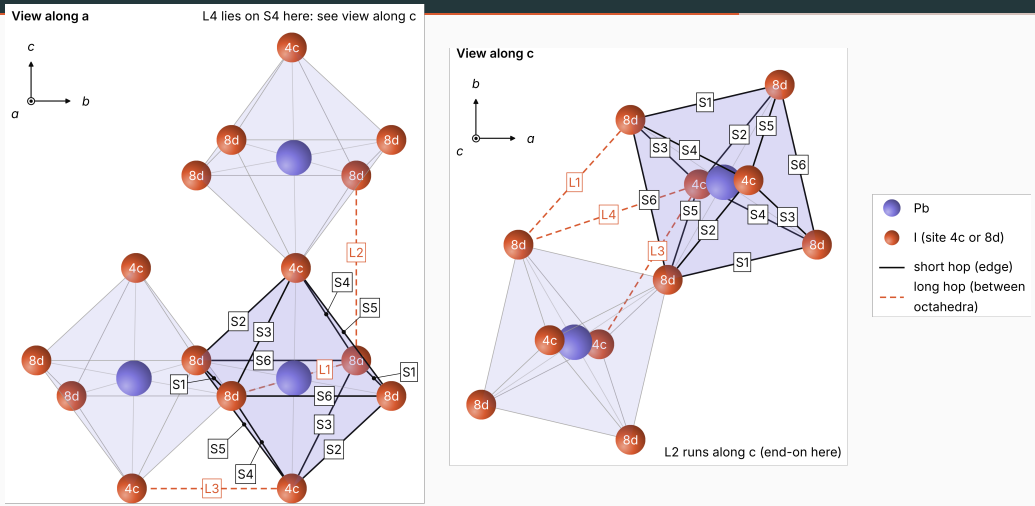
Why do we disagree with paper A? The relaxed lattice

Source	Method	Short / long in-plane axis (Å)	c (Å)	In-plane difference
Experiment, 293 K[4]	X-ray	8.577 / 8.856	12.47	3.3%
Kaczkowski 2021[5]	LDA	7.947 / 8.939	12.22	12.5%
This work	**PBEsol**	**8.362 / 8.961**	**12.35**	**7.2%**
Zhang 2025[20]	PBEsol	8.39 / 8.96	12.37	6.8%
Zhang 2025[20]	HSE06	8.40 / 9.03	12.59	7.5%
Seidu 2021[6]	PBEsol	8.31 / 8.94	12.35	7.6%
Kaczkowski 2021[5]	SCAN	8.522 / 9.018	12.52	5.8%
Kaczkowski 2021[5]	PBE	8.731 / 9.095	12.64	4.2%
Paper A[1]	PBEsol	8.68 / 8.71	12.44	0.3%

- Every 0 K relaxation is clearly rectangular, *including a hybrid functional*; ours matches two independent PBEsol cells.
- Paper A's cell is nearly square ⇒ weaker tilt ⇒ 4c and 8d more alike.
- Since the sites differ only through tilt, this may explain part of the disagreement.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [4] R. J. Sutton *et al.*, ACS Energy Lett. 3, 1787 (2018). [5] J. Kaczkowski, I. Płowaś-Korus, J. Phys. Chem. Lett. 12, 6613 (2021). [6] A. Seidu *et al.*, J. Chem. Phys. 154, 074712 (2021). (converted by us from its rotated 2×2×2 setting). [20] J. Zhang *et al.*, Appl. Surf. Sci. 679, 161235 (2025). (Table S1).

RQ2 set-up: ten distinct hops, not three



S1–S6: along octahedron edges; L1–L4: between corner-sharing octahedra. Figure, hop set and lengths: this work (tight-relaxed cell).

The ten hops: how many, and where they go

Hop	Ends	Length (Å)	From 8d	From 4c	Moves the vacancy
S1, S6	8d–8d	4.41, 4.60	2 each	0	within a PbI_2 layer
S2–S5	4c–8d	4.43–4.56	1 each	2 each	$PbI_2 \leftrightarrow$ CsI layer
L1	8d–8d	4.80	1	0	within a PbI_2 layer
L2	8d–8d	5.18	1	0	straight along c
L3	4c–4c	5.28	0	2	within a CsI layer
L4	4c–8d	5.93	1	2	between layers
Total			11	12	

- Half of the short hops from 8d lead to 4c; a 4c vacancy has **no short hop to another 4c**
- ab plane: 8d–8d hops are enough. Along c: needs **4c–8d hops** (8d→4c→8d) or the long L2
- ⇒ transport depends on barriers *and* hop numbers *and* 4c/8d site energies: anisotropic diffusion

Hop set, counts and directions: this work (`hop_network.py`, perfect supercell). Paper A treats one barrier per type[1]. [1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025).

NEB scope: start with hops that have published values

Published path	Value	Our hop	How identified
Paper A: 4c–4c[1]	0.77 eV	**L3**	only 4c–4c hop
Paper A: 8d–8d edge[1]	0.34 eV	S1 or S6	schematic figure only
Paper A: 8d–4c edge[1]	0.35 eV	one of S2–S5	schematic figure only
Zhang: in-plane[20]	0.54 eV	**S1** (probable)	rises ≈1 Å along c (S6: 0)
Zhang: out-of-plane[20]	0.51 eV	**S2** (probable)	rises ≈2.7 Å, no sideways shift

Stage	Hops	Purpose
1	S1, S2, L3, S6	Compare with published values; S1 vs S6 = first spread within a type
2	S3, S4, S5	Spread of the 4c–8d hops (control transport along c)
3	L1, L2, L4	Only if the network analysis needs them

Spread < 0.03 eV $\Rightarrow$ one barrier per type. Small threshold: at 300 K, 0.05 eV changes a rate $\approx 7\times$.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [20] J. Zhang *et al.*, Appl. Surf. Sci. 679, 161235 (2025). (paths assigned from its Fig. 6a,b; raster figure, ± 0.3 Å).

Next steps

NEB: ready to run

`build_neb.py` (written and tested)

- End points by symmetry: the relaxed 4c and 8d vacancies are mapped onto any site with the 64 Pnma operations[17] → no extra relaxations
- Atoms renumbered so only the hopping iodine moves; IDPP starting path[16,18]

NEB settings and why

Climbing image	on	Gives the true saddle point[15]
Intermediate images	4	Spacing about 1 Å; the climbing image finds the saddle
k-mesh	2×2×2	Same as the end points (2×2×1 rejected)
Barrier reported	both directions	Plus the average, to compare with paper A[1]

First run: S1 (8d–8d, +1), against paper A's 0.34 eV.

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025). [15] G. Henkelman, B. P. Uberuaga, H. Jónsson, J. Chem. Phys. 113, 9901 (2000). [16] S. Smidstrup *et al.*, J. Chem. Phys. 140, 214106 (2014). [17] A. Togo, K. Shinohara, I. Tanaka, Sci. Technol. Adv. Mater.: Methods 4, 2384822 (2024). [18] A. H. Larsen *et al.*, J. Phys.: Condens. Matter 29, 273002 (2017).

Plan

1. Finish the neutral 4c relaxation → final neutral site energy
2. NEB stage 1 (+1): S1, S2, L3, S6 (published comparisons); stage 2: S3, S4, S5
3. Long hops L1, L2, L4 only if the network analysis needs them
4. Neutral NEB for the stage-1 hops
5. Build the CsI-terminated (001) slab: symmetric, all atoms free; 11 layers, 15-layer test
6. Vacancy energies and barriers layer by layer, until they return to the bulk values

Questions for discussion

1. **Paper A's lattice**[1]**:** rerun our +1 pair on their cell (8.68/8.71/12.44 Å) to test the tilt explanation? Cost: two relaxations, about one week.
2. **Neutral vacancy:** is a hybrid-functional check of the delocalised electron worth the cost?
3. **Spin–orbit coupling:** single-point check on the site energies now, or after the barriers?
4. **NEB scope:** start with the 4 hops comparable to published values, then decide; neutral only for stage 1?
5. Contact paper A's authors about their charge state and 4c/8d end-point energies?

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025).

Summary

- **Bulk reference built:** converged settings, tight lattice, perfect and vacancy supercells.
- **Site preference depends on charge:** $+1$ degenerate (-1 meV); neutral favours 4c by ≥ 36 meV, opposite to paper A.
- **Neutral vacancy is a shallow donor** in PBEsol (moment $0\,\mu_B$).
- **Benchmark mismatch** likely from paper A's nearly square, less-tilted cell.
- **Ten distinct hops**; transport along c needs 4c–8d hops $\Rightarrow$ staged NEB plan, inputs ready.

Next: first barrier (S1) against paper A's 0.34 eV.

References i

[1] A. N. Arber, Vikram, F. C. Mocanu, M. S. Islam, Chem. Mater. 37, 4416 (2025).
[2] R.-I. Biega, L. Leppert, J. Phys.: Energy 3, 034017 (2021).
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